Analysis Qualifier Prep
Problem Set 1 (Rookie Mix)

Uttaran Choudhurry

June 2026

Question 1:

Determine, with justification, the limit

$$\lim_{n\to\infty} \int_0^\infty \frac{1 - cos(nx)}{n^2 x^2} dx$$

Question 2:

Let (X, M, μ) be a finite measure space. Let $f : X \to \mathbb{C}$ be a measurable function. Show that $f \in L^\infty(X, \mu)$ if and only if

$$\sup_p \Big\{ \|f\|_p : p \in [1, \infty) \Big\} < \infty$$

Question 3:

Let $f \in L_1([0, \infty))$ (with respect to the usual Lebesgue measure). Suppose that f is uniformly continuous. Prove that $f \in C_0([0, \infty))$.

Question 4:

Let $C([0, 1])$ be the space of continuous functions on $[0, 1]$. Prove that

$$\|f\| = \left(\int_{[0,1]} |f|^2 dm \right)^{1/2}$$

defines a norm on $C([0, 1])$. Is $C([0, 1])$ a Banach space with respect to this norm? Justify your answer.

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Problem Set 2

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Question 1:

Suppose f is a Lebesgue measurable function on $\mathbb{R}$ such that f and $x :\to xf(x)$ are both in $L^2(\mathbb{R})$. Prove that $f \in L^1(\mathbb{R})$

Question 2:

Suppose that $f : [1, \infty) \to \mathbb{C}$ is Lebesgue integrable and f is continuous at 1. Prove that

$$\lim_{n\to\infty} n \int_1^\infty \frac{f(x)}{x^n} dx = f(1).$$

Question 3:

Let $p \in (1, \infty)$ and let $f_1, f_2, ... : \mathbb{R} \to \mathbb{C}$ be L^p functions. Assume that $\sum_{n=1}^\infty \|f_n\|_p^p < \infty$.

1. Suppose that the sets $E_n = \{x \in \mathbb{R} : f_n(x) \neq 0\}$ are disjoint. Prove that the series $\sum_{n=1}^\infty f_n$ converges in $L^p(\mathbb{R})$.

2. Give an example to show that the conclusion of part (1) can fail if the sets E_n are not assumed disjoint.

Question 4:

For which $p \in [1, \infty]$ is there a nonzero bounded linear functional T on $L^p([0, 1])$ such that T vanishes on the subspace $C([0, 1]) \subset L^p([0, 1])$? Justify your answer.

Analysis Qualifier Prep
Problem Set 3

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Question 1:

Let $\{f_n\}$ be a sequence of Lebesgue integrable functions on the real line. Suppose that $|f_n(x)| \leq 1$ for all $x \in \mathbf{R}$ and f_n converges to zero almost everywhere. Is it true that

$$\lim_{n\to\infty} \int_{\mathbf{R}} f_n dm = 0,$$

where m is the Lebesgue measure? It true explain else if false, explain what had to change.

Question 2:

Let $(X, \mathcal{M}, \mu)$ be a finite measure space and let $\{f_n\}$ be a sequence of measurable functions which converges to zero almost everywhere (with respect to μ). Prove that f_n converges to zero in measure.

Question 3:

Let $(f_n)_{n\in\mathbb{N}}$ be a sequence of functions in $L^2(\mathbb{R})$. Suppose that for all $n \in \mathbb{N}$, the Fourier transform $\widehat{f_n}$ vanishes outside $[n, n+1]$ and that

$$\|f_n\|_2 = n^{-\frac{51}{100}}.$$

Show that the series $\sum_{n=1}^{\infty} f_n$ converges in norm in $L^2(\mathbb{R})$.

Question 4:

Let $f \in L^1(\mathbb{R})$ and $g = \chi_{[0,1]}$. Suppose that their convolution $f * g = 0$ for a.e $x \in \mathbb{R}$. Prove that $f = 0$ a.e.

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Problem Set 4

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Question 1:

Recall that l^∞ is the set of bounded sequences of real numbers. Define $c = \{\{x_n\} \in l^\infty, \lim_{n\to\infty} x_n \text{ exists}\}$ with the induced supremum norm. Show

(i) For $\{\eta_n\} \in l^1$ and $\alpha \in \mathbb{R}$, functional

$$f(\{x_n\}) = \sum_{n=1}^{\infty} \eta_n x_n + \alpha \cdot \lim_{m\to\infty} x_m$$

is a bounded linear functional (i.e., in $(c)^*$).

(ii) The norm of the functional f is given by

$$\|f\| = \|\{\eta_n\}\|_{l^1} + |\alpha|.$$

Question 2:

Let $T : X \to Y$ be bounded linear operator between normed spaces.

(a)Prove that for any $x, w \in X$ we have

$$\max\{\|T(x+w)\|, \|T(x-w)\|\} \geq \|T(x)\|. \tag{1}$$

(b)Prove that for any $x \in X$ and $r > 0$

$$\sup_{\tilde{x}\in B(x,r)} \|T(\tilde{x})\| \geq r\|T\|. \tag{2}$$

where $B(x, r)$ is the open ball in X centered at x with radius r.

Question 3:

Suppose that $\{f_n\}$ is a sequence of Lebesgue integrable function on $[0, 1]$ such that $|f_n(x)| \leq \pi$ for all n and $x \in [0, 1]$. Suppose that

$$\lim_{n\to\infty} f_n(x) = 0 \text{ } a.e.m. \text{ and } \left|\frac{f_n(x)}{x^2}\right| \leq x^{1/2} \text{ for all } x \in (0, 1], \text{ } a.e.m. \text{ and } n \in \mathbb{N}.$$

Compute the

$$\lim_{n\to\infty} \int_{(0,1]} \frac{f_n(x)}{1-\cos(x)} dm,$$

where m is the Lebesgue measure. Justify your computation.

Question 4:

Give an example of a sequence of functions with $\inf_n \|f_n\|_\infty \geq M$ for some $M > 0$ and a function f such that $\lim_{n\to\infty} \|f_n - f\|_p = 0$ for all $p > 0$, but for which $\|f\|_\infty < M$.

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Problem Set 5

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Question 1:

Let μ be a positive measure on X. Prove that

$$0 < r < p < s < \infty \implies L^r(\mu) \cap L^s(\mu) \subset L^p(\mu).$$

Question 2:

Find a bounded linear functional L on $L^p(\mathbb{R})$, $1 \leq p < \infty$, such that

$$L(\chi_{[k,k+1]}) = (1 + |k|)^{-1} \quad \text{for all } k \in \mathbb{Z}.$$

Question 3:

Let $X = \mathbb{R}$ be equipped with the standard Borel σ-algebra $\mathcal{B}$. Let m be the Lebesgue measure on $\mathbb{R}$, and let δ_x denote the Dirac measure concentrated at x.

Define two new measures on $(X, \mathcal{B})$:

$$\mu = m + \sum_{n=1}^{\infty} \frac{1}{2^n} \delta_n$$

$$\nu = m$$

1. Find the Lebesgue decomposition of μ with respect to ν.
2. Find the Lebesgue decomposition of ν with respect to μ. Does the Radon-Nikodym derivative $\frac{d\nu}{d\mu}$ exist? If so, compute it.
3. For $1 \leq p < \infty$, give a precise description of the functions that belong to the space $L^p(\mu)$.

Question 4:

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Problem Set 6 (Practice)

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Question 1:

Find the limit

$$\lim_{n\to\infty} \int_{\frac{1}{n}}^{n} e^{\frac{1}{n}\sqrt{x}-x}$$

Question 2:

Compute

$$\lim_{n\to\infty} \iint_{(0,\infty)^2} \frac{n}{x} \sin\left(\frac{x}{ny}\right) e^{-\frac{x}{y}-y}\,(dx\,dy).$$

Make sure to fully justify your computations.

Question 3:

Prove that a linear functional f on a normed linear space X is bounded if and only if the kernel of f, i.e.

$$\ker f = \{x \in X : f(x) = 0\},$$

is closed.

Question 4:

Let m denote Lebesgue measure on $\mathbf{R}$. Recall that for any measurable set $E \subset \mathbf{R}$ with $m(E) > 0$, and any $\epsilon > 0$, there exists a finite interval $I \subset \mathbf{R}$ such that $m(I \cap E) > (1-\epsilon)m(I)$. Use this to prove that if $E, F \subset \mathbf{R}$ with $m(E) > 0$ and $m(F) > 0$ then the set

$$E - F = \{x - y : x \in E, y \in F\}$$

contains a nontrivial interval.

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Problem Set 7

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Question 1:

Compare $\int_X f d\mu$ and $\sum_{n=1}^{N} \mu\{x : f(x) \geq n\}$

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Problem Set 8

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Question 1:

(a) For $f \in L^2(\mathbb{R})$ and a sequence $\{x_n\} \subseteq \mathbb{R}$ which converges to zero, define $f_n(x) := f(x + x_n)$. Show that $\{f_n\}$ converges to f in L^2.

(b) Let $W \subseteq \mathbb{R}$ be a Lebesgue measurable set of positive Lebesgue measure. Show that the set of differences $W - W = \{x - y : x, y \in W\}$ contains an open neighborhood of the origin.

Question 2:

Let $f(x)$ be a non-decreasing function on $[0, 1]$.

(a) Prove that $\int_0^1 f'(x)dx \leq f(1) - f(0)$.

(b) Let $\{f_n\}$ be a sequence of non-decreasing functions on $[0, 1]$ such that the series $F(x) = \sum_{n=1}^{\infty} f_n(x)$ converges for all $x \in [0, 1]$. Prove that $F'(x) = \sum_{n=1}^{\infty} f_n'(x)$ almost everywhere.

Question 3:

Let $I_{0,0} = [0, 1]$ and for $n \geq 0$, $0 \leq j \leq 2^n - 1$, let

$$I_{n,j} = [\frac{j}{2^n}, \frac{j+1}{2^n}].$$

For $f \in L^1([0, 1])$ define $E_n f = \sum_{j=0}^{2^n-1} \left(2^n \int_{I_{n,j}} f(t)dt\right) \chi_{I_{n,j}}$. Prove that $E_n f \to f$ almost everywhere on $[0, 1]$.